

MATERIAL SAFETY DATA SHEET

1. Product and company identification

Product name : 2-HYDROXYPROPYL ACRYLATE

Name of supplier	NIPPON SHOKUBAI CO., LTD.
Address	Kogin Bldg., 4-1-1 Koraibashi, Chuo-ku, Osaka 541-0043 Japan
Tel. number	81-6-6223-9166
Fax. number	81-6-6202-1766
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Emergency Tel. number	81-79-273-1139 Fax. number 81-79-274-3723 (81:Country No. of Japan)
Date PREPARED	3. Aug. 1996
Date REVISED	1. Dec. 2010

• Recommended use of the chemical and restrictions on use: coating, adhesive

MSDS-No. HPI-1212-00-10

2. Hazards identification

• The most important hazards:

GHS classification: Flammable liquids	not classified
Pyrophoric liquids	not classified ⁶⁾
Acute toxicity(oral)	category 4 ⁶⁾
Acute toxicity(dermal)	category 2 ⁶⁾
Skin corrosion/irritation	category 1A – 1C ⁶⁾
Serious eye damage/eye irritation	category 1 ⁶⁾
Respiratory or skin sensitization	category 1 ⁶⁾ (skin sensitization)
Specific target organ systemic toxicity (Single exposure)	category 2 ⁶⁾ (respiratory system)
Specific target organ systemic toxicity (Repeated exposure)	category 1 ⁶⁾ (respiratory system)
Hazardous to the aquatic environment (Acute)	category 2 ⁶⁾
Hazardous to the aquatic environment (Chronic)	not classified ⁶⁾

GHS label elements

Pictograms or symbols:



Signal word : Danger

Hazard statement : Harmful if swallowed.

Fatal in contact with skin.

Causes severe skin burns and eye damage.

Causes serious eye damage.

May cause an allergic skin reaction.

May cause damage to organs (respiratory system) .

Causes damage to organs(respiratory system) through prolonged or repeated exposure.

Toxic to aquatic life.

Precautionary statements:

[Prevention]

- Do not eat, drink or smoke when using this product.
- Wash thoroughly after handling.
- Do not get in eyes, on skin, or on clothing.
- Wear protective gloves/clothing and eye/face protection.
- Do not breathe dust/fume/gas/mist/vapors/spray.
- Contaminated work clothing should not be allowed out of the workplace.
- Avoid release to the environment.

[Response]

- IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
- Rinse mouth.
- Remove/Take off immediately all contaminated clothing.
- IF ON SKIN: Gently wash with plenty of soap and water.
- Immediately call a POISON CENTRE or doctor/physician.
- Specific treatment is urgent.
- Wash/Decontaminate removed clothing before reuse.
- Rinse skin with water/shower.
- Wash contaminated clothing before reuse.
- IF IN EYES: Rinse cautiously with water for several minutes.
Remove contact lenses, if present and easy to do. Continue rinsing.
- IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
- IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing.
- If skin irritation or rash occurs, seek medical advice/attention.
- Get medical attention/advice if you feel unwell.

[Storage]

- Store locked up.

[Disposal]

- Dispose of contents/container to consult an expert in accordance with local/regional/national/international regulation.

Other hazards which do not result in classification by the GHS :

Although polymerization inhibitors are added, polymerization may be induced by heat, sunlight, peroxides, iron rust, etc.

If the polymerization takes place rapidly, temperature rises rapidly, and the accelerated rise of vapor pressure may lead to an explosion.

3. Composition/information on ingredients

- Substance/mixture: : mixture (Inhibited)

• Chemical name	2-Hydroxypropyl acrylate	2-Hydroxy-1-methylethyl acrylate
• Chemical formula	$\text{CH}_2=\text{CHCOOCH}_2\text{CH}(\text{OH})\text{CH}_3$	$\text{CH}_2=\text{CHCOOCH}(\text{CH}_3)\text{CH}_2\text{OH}$
• Concentration range	70~80%	20~30%
	98~99%	
• CAS number	25584-83-2	

4. First-aid measures

- Inhalation : Remove the victim to fresh air immediately. Keep the victim warm and rest. Get immediate medical attention.
- Skin contact : Immediately flush skin with plenty of water. Immediately take off contaminated clothing and flush skin with plenty of water. Get medical attention immediately.
- Eye contact : Immediately flush eyes with plenty of clean water for at least 15 minutes (remove contact lenses if easily possible). Immediately get medical attention. Keep enough opening the eyelid while flushing. Wash every corner of eye.
- Ingestion : Do not induce vomiting. Immediately give the victim one or two glasses of milk or water. Immediately call a physician. Never give anything by mouth to an unconscious person.

5. Fire-fighting measures

- Suitable extinguishing media:
Use powder, alcohol-resistant foam, carbon dioxide.
- NOT suitable extinguishing media: Not available
- Specific methods of fire-fighting:
Fight fire from windward. Shut off fuel to fire and use extinguishing agents.

If the fire spread around, apply water to cool and protect surrounding area.

- Special equipment for the protection of firefighters:

As in any fire, wear self-contained breathing apparatus with full face piece operated in positive pressure mode and full protective gear.

6. Accidental release measures

- Personal precautions :

Evacuate personnel from the lee. Evacuate non-essential personnel. Shut off all sources of ignition; No flares, smoking, or flames in the area. Wear proper protective equipment.

- Environmental precautions :

Do not flush to sewer or waterways. Dike with soil. Cover with a sheet to prevent from expanding odor. Because of unpleasant odor, you manage appropriately such that you inform the fact to the resident that leakage occurred.

- Methods for cleaning up :

For small spills, use caustic soda solution of 5 to 10% for hydrolysis and neutralization then wash with water. Process the drainage appropriately (incineration, activated sludge process). For large spills, enclose the spilled liquid with sand. Recover the liquid while covering it with an oil-resistant antistatic sheet to suppress the evaporation of its vapor and treat this substance with the same way of small spills.

7. Handling and storage

Handling

- Appropriate technical measures:

Do not breathe vapor. Avoid contact with eyes, skin and clothing.

Wear adequate protective equipment. Operate from windward.

After operation, lock up the container.

Prevent build-up of electrostatic charge (e.g. by grounding).

Prevent the human body electrification by wearing electrostatic prevention clothes and electrostatic prevention shoes.

Use in the local exhaust ventilation.

- Precautions for safe handling:

Don't leak, overflow, and scatter. Minimize vapor generation and accumulation.

Shut off all sources of ignition; No flares, flames or high-temperature substance in the area. Do not place close to strong oxidizing materials or peroxides that may cause inflammation or polymerization by contact or mixing. Avoid falling, dropping, shocking and dragging a container.

- Specific safe handling advice: Not available

Storage

- Suitable storage conditions :

When it stores in a tank, keep the liquid temperature not exceeding 40°C and the gas phase concentration of oxygen from 7 to 21 vol%. (In the presence of oxygen under 7 vol%, polymerization may occur.)

The electrical apparatus makes explosion-proof construction. The tank and apparatus are grounded.

Do not store in direct sunlight. Keep away from heat and flame.

- Safe packaging materials:

Packaging materials which can be sealed. Stainless steel or polyethylene is suitable.

8. Exposure controls/personal protection

- Engineering measures :

Facilities storing or utilizing this substance should be equipped with an eyewash facility and a safety shower. Local exhaust ventilation may be necessary to control any air contaminants to within their TLVs during the use of this material.

- Control parameters :

[as 2-Hydroxypropyl acrylate (CAS No. 999-61-1)]

ACGIH(2009) TWA 0.5ppm (skin, sensitization)

OSHA(2009) Not established

- Personal protective equipment :

Respiratory protection: Chemical cartridge respirator with an organic vapor cartridge, supplied-air respirator, or self-contained breathing apparatus.

Hand protection: Chemical resistant gloves

Eye protection: Safety glasses with side shields, goggles, a face shield

Skin and body protection : Suitable safety clothes and shoes

9. Physical and chemical properties

- Appearance

Form: Liquid

Color: Colorless

- Odor: ester odor

- Odor threshold: Not available

- pH: 3.4 (20°C 1mol/L aqueous solution)¹⁾

- Boiling point: 205.7 °C³⁾

- Initial boiling point and boiling range: Not available

- Melting point: -23.35 °C³⁾

- Decomposition temperature: Not available

- Flash point: : 109 °C (Cleveland open cup)²⁾
- Auto-ignition temperature: Not available
- Flammability: Not available
- Explosion properties
 - Explosive limits: lower explosive limit 1.8% (v/v)⁴⁾
 - Dust explosion: Not applicable
- Vapor pressure: 0.038 hPa (25°C)³⁾
- Vapor density: Not available
- Rate of vaporization: Not available
- Density/Specific gravity: 1.049 g/cm³ (25°C)³⁾
- Solubility
 - In water: Very good¹⁾
- Octanol/water partition coefficient: 0.35 (log Kow)³⁾
- Other data: Not available

10. Stability and reactivity

- Stability: Although polymerization inhibitors are added, polymerization may be induced by heat, sunlight, peroxides, iron rust, etc.
- Possible hazardous reactions:
 - If stored in the presence of oxygen under 7 vol%, polymerization may occur.
- Conditions to avoid: overheat, contact with materials to avoid and fire
- Materials to avoid: strong oxidizing agents and peroxide
- Hazardous decomposition products: not applicable

11. Toxicological information

- Acute toxicity:

Oral	rat (male)	LD ₅₀	570 mg/kg ²⁾
	rat (female)	LD ₅₀	510 mg/kg ²⁾
	rat	LD ₅₀	820 mg/kg ³⁾
dermal	rabbit	LD ₅₀	306 mg/kg ³⁾
- Local effects
 - Skin corrosion/irritation: No corrosion (as this product)
Severely irritating (as this product)
(Tested by OECD Test Guideline[404])²⁾
 - Rabbit Highly irritating³⁾
 - human Irritating³⁾
 - Serious eye damage/eye irritation:
 - Rabbit May cause severe irritation with corneal injury
which may result in permanent impairment of

vision, even blindness³⁾

• Sensitization

Skin sensitization	guinea pig	Not sensitizing - Sensitizing ³⁾
	Human	Sensitizing ³⁾

• Germ cell mutagenicity:

In Vitro:

Reverse mutation assay	Salmonella typhimurium	Negative (with and without metabolic activation) ³⁾
Reverse mutation assay	E. coli	Positive with metabolic activation ³⁾
Cytogenetic assay	Chinese hamster	Positive (with and without metabolic activation) ³⁾
Gene mutation assays	Chinese hamster	Negative (with and without metabolic activation) ³⁾
Chromosome aberration	CHO cells	Positive (with and without metabolic activation) ³⁾

In Vivo: mouse micronucleus

Negative ³⁾

• Carcinogenicity : Not established on IARC, NTP, EU, OSHA, ACGIH.

• Reproductive toxicity:

inhalation rat, mouse, rabbit, dog No effects at 5 or 10 ppm (4-week males) ³⁾

Teratogenicity:

Inhalation	rat	NOAEL maternal tox > 1ppm (5.3 mg/m ³) ³⁾
Inhalation	rat	NOAEL fetotoxicity > 10ppm (53 mg/m ³) ³⁾
Inhalation	rat	NOAEL teratogenicity > 10ppm (53 mg/m ³) ³⁾

• Chronic toxicity or long term toxicity

Single dose toxicity No information regarding (swallow/skin absorption/inhalation) toxicity data.

Repeated dose Toxicity

inhalation	rat and mice	NOAEC 5 ppm (27 mg/m ³) ³⁾
inhalation	rat and mice	LOAEC 10 ppm (53 mg/m ³) ³⁾
inhalation	rabbits and dogs	LOAEC 50 ppm (27 mg/m ³) ³⁾

• Aspiration hazard : No relevant information found.

12. Ecological information

• Ecotoxicity

Fish :	Pimephales	LC ₅₀ (96h)	3.1 mg/L ³⁾
	Promelas	EC ₅₀ (96h)	1.9 mg/L ³⁾
Crustacea :	Daphnia magna	EC ₅₀ (48h)	24 mg/L ³⁾
	Daphnia magna	NOEC ₅₀ (48h)	10 mg/L ³⁾

Algae or other aquatic plants:

Algae	E _b C ₅₀ (72h)	3.88mg/ L ³⁾
Algae	E _b C ₅₀ (96h)	3.53mg/ L ³⁾
Algae	E _r C ₅₀ (72h)	6.98mg/ L ³⁾
Algae	E _r C ₅₀ (96h)	6.67mg/ L ³⁾

- Persistence/degradability : This substance is biodegradable. ⁵⁾
- Bioaccumulation : Not available
- Mobility : It is possible to transfer to the atmosphere, the hydrosphere and the soil.

13. Disposal considerations

- Waste from residues:
Burn in a chemical incinerator equipped with an afterburner and scrubber. Consult an expert on the disposal of recovered material.
- Any contaminated packaging:
Do not put other material into the used container and do not use it for other purpose. Treat the contents burned on the above. Wash the inside of the container before disposal. Recycling drums should be completely drained, properly bunged and promptly returned to a drum reconditioner.
- Comply with all federal, state and local regulations.
- Do not dump this product into sewers, on the ground or into any body of water.

14. Transport information

- The UN classification number
[IMDG]
Proper shipping name : TOXIC LIQUID ORGANIC, N. O. S
UN class : 6.1(toxic)
UN number : 2810
Packing group : III
Marine pollutant: not applicable
- Specific precautionary transport measures and conditions :
Avoid falling, dropping, shocking and dragging a container. Protect a container from direct sunlight. Secure the grounding of the vehicle to avoid static electrification before starting the liquid transfer. After discharging the product from a tank lorry, the remnant liquid in piping must be completely eliminated.
- Any transportation practice must be in compliance with laws and regulation in your country or region.

15. Regulatory information

	as Products (2-Hydroxypropyl acrylate) (CAS No. 25584-83-2)
US STATUS	
OSHA STATUS	Not listed
TSCA inventory	Listed
SARA TITLE III 313	Not listed
EU STATUS	
REACH Registration	01-2119459351-41
REACH Restriction (EC No 1907/2006)	Not listed in Annex XIV.
Classification & labeling	
(EC) No 1272/2008 (CLP) [CLASSIFICATION]	Acute Tox. 3 * H331 Acute Tox. 3 * H311 Acute Tox. 3 * H301 Skin Corr. 1B H314 Skin Sens. 1 H317
[LABELLING]	GHS06 H331 GHS05 H311 Dgr H301 H314 H317
(67/548/EEC) [CLASSIFICATION]	T;R23/24/25 C;R34 R43

• Registration STATUS in the lists of existing chemicals of various countries

	as Products (2-Hydroxypropyl acrylate) (CAS No. 25584-83-2)
Canada (CEPA)	Listed in DSL
Japan (ENCS)	(2)-997
Australia (AICS)	Listed
Korea (ECL)	KE-29612
Switzerland (SWISS)	(EINECS) 247-118-0
Philippine (PICCS)	Listed
China (SEPA)	Listed

• Restriction at export: Not applicable

- Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.

16. Other information

- References:
 - 1) Data generated by NIPPON SHOKUBAI CO., LTD.
 - 2) Data generated by external institute at a request of NIPPON SHOKUBAI CO., LTD.
 - 3) OECD/HPV (SIAM20 19-22 Apr. 2005)
 - 4) G. Weiss, "Hazardous Chemical Data Book", Second Edition.
 - 5) The Official Bulletin of Ministry of Economy, Trade and Industry (10. May. 2001)
 - 6) National Institute of Technology and Evaluation (NITE) :
GHS classification database
- To the best of our knowledge, the information contained here in is accurate. However, neither NIPPON SHOKUBAI CO., LTD. nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards which exist.

ORIGINAL CONTROLLED